

GROUP III – CONCEPTUAL QUESTIONS

Student: Tomás Carvalho | N:16 | Course: TEARCI T3

C1. LM35 Operating Principle

The LM35 is a temperature sensor that produces an output voltage that changes according to the temperature. Inside the IC, there are two transistors working at different current levels, and the difference between their V_{BE} voltages creates a signal that varies proportionally with the absolute temperature (PTAT).

The internal circuitry then converts this PTAT signal into a voltage that corresponds directly to degrees Celsius. After calibration, the LM35 provides 10 mV for each °C, which makes the measurement very simple and linear.

C2. CA3130 Comparator and Reference Network

In this project, the CA3130 is used as a comparator in open-loop mode. Its role is to check if the voltage from the LM35 (which represents the temperature) is above or below a set threshold.

The non-inverting input (+) receives the LM35 voltage.

The inverting input (–) receives the reference voltage (V_{ref}), which is set through the R3–P1 divider.

The output works digitally:

HIGH when the measured temperature is above the setpoint.

LOW when it is below.

By adjusting the potentiometer (P1), we can define the exact temperature where the comparator should switch. This is what determines when the alarm activates.

C3. Cascaded NPN Buffer/Amplifier

This stage is necessary because the photodiode produces a very weak current, in the microamp range, which is not enough to drive the next IC. By using two NPN transistors in cascade, the signal is amplified and cleaned so it becomes a proper square wave.

Some important advantages of this configuration:

High gain: allows the circuit to detect even low IR levels.

Noise reduction: the second transistor saturates and clips unwanted noise.

Pulse shaping: the transitions become sharper, generating a stable square-wave signal for the 555/567.

C4. 555 Astable Tone Generation

The NE555 is configured in astable mode to generate the tone for the buzzer. Since the buzzer needs an alternating signal to vibrate, the 555 produces a square wave that makes the buzzer emit a continuous sound.

The output frequency depends on R7, R8, and capacitor C4. Adjusting these components changes the tone. When the frequency is close to the buzzer's resonant point, the sound becomes louder and more efficient.

This continuous tone is easier to notice than a simple click, which makes it suitable for alarms.

Sensor Simulation – LM35

To simulate temperatures like 25°C, 50°C and 75°C, we adjusted the LM35 output. The voltage increased according to the sensor's scale (10 mV/°C). In the simulator, the values approached around 4.24 V, depending on the configuration.

3) Comparator Output Observation

Low state (normal condition):

The output stayed close to 0 V.

High state (temperature above threshold):

When the LM35 voltage passed the reference value, the comparator switched to about 5 V.

Signal behavior:

The transition was clean and didn't show oscillations.

Since the LM35 output rises 10 mV per °C, the comparator threshold directly defines the temperature where the alarm activates.

4) Tone Generator (NE555) Testing:

Output signal (pin 3):

A square wave between 0 V and 5 V.

Frequency:

Close to 400 Hz, depending on R7, R8 and C4.

Buzzer behavior:

The buzzer emitted a continuous tone matching the 555 frequencies.

Component adjustments:

Lower R or C → higher frequency

Higher R or C → lower frequency

This matched the astable configuration expected from the NE555.



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